Mathematics AI HL · Set C · Paper 3

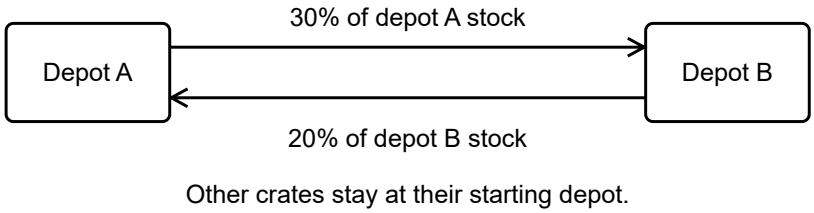
Question booklet · 75 minutes · 55 marks

Answer all questions. A calculator is permitted. Show your reasoning. Give exact answers or round to three significant figures unless directed otherwise. Write on separate paper.

1. Investigation 1 [28]

Redistributing a shared fleet

Six hundred reusable crates circulate between depots A and B. Each evening 30% of the crates at A go to B and 20% of those at B go to A. The rest stay. Let x_n be the expected number at A after n evenings. Initially $x_0 = 480$. Use a column state vector in the order A, B.



Other crates stay at their starting depot.

(a) Write the transition matrix and find the expected numbers at both depots after one evening. **[3]**

(b) Show that $x_{n+1} = 0.5x_n + 120$. Explain why the constant is independent of n. **[3]**

(c) Find the equilibrium allocation. Use a deviation from equilibrium to obtain a formula for x_n. **[4]**

(d) Determine the first evening on which A is within five crates of equilibrium. Show that your answer is the first. **[4]**

(e) The manager claims that if the fleet starts at any allocation, A will eventually have exactly 240 crates in expectation. Evaluate the claim. **[3]**

(f) A van now returns s extra crates from B to A after each ordinary transfer. Find the least whole s for which A's long-run expected stock is at least 300. Check availability at equilibrium. **[5]**

(g) Operating the van costs 18 units per crate moved per evening. A shortage below 300 at A costs 50 units per missing crate per evening. For whole $0 \le s \le 60$, determine the long-run cost-minimizing policy under this expected-stock model. Explain one limitation of the cost model. **[6]**

2. Investigation 2 [27]

Water storage through a tide

A collection tank starts with 18 cubic metres of water. From $t = 0$ to 12 hours its inflow is $6 + 4\sin(\pi t/6)$ cubic metres per hour, and a pump removes 7 cubic metres per hour. Begin with an unlimited tank. Use radians.

(a) Find the net rate at $t = 0$ and at $t = 3$. **[2]**

(b) Find an expression for the stored volume $V(t)$ and verify the initial condition. **[4]**

(c) Determine when the unlimited tank reaches its greatest volume and find this volume. Justify your choice on the whole interval. **[4]**

(d) The actual tank holds 25 cubic metres. Find when spillage first occurs and how much is spilled before inflow next falls below pump output. **[4]**

(e) Find the actual volume at 12 hours. Explain whether the pump runs dry at any time. **[4]**

(f) The operator can change the starting volume while leaving both rates unchanged. Find the full interval of starting volumes that avoid overflow and keep the tank from emptying throughout the 12 hours. Touching full capacity is allowed. **[5]**

(g) A technician estimates volume change using readings of net rate at $t = 0, 6, 12$ and two trapezia. Compare the estimate with the true net change, and judge whether those readings can certify that no spill occurred. **[4]**

End of question booklet · 55 marks